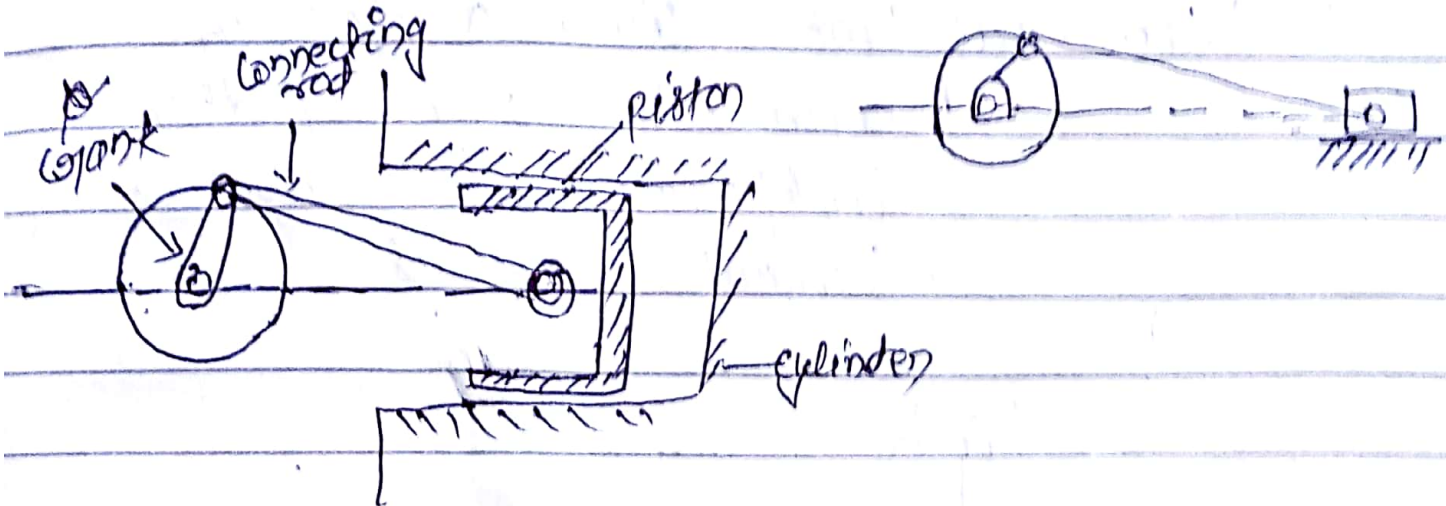
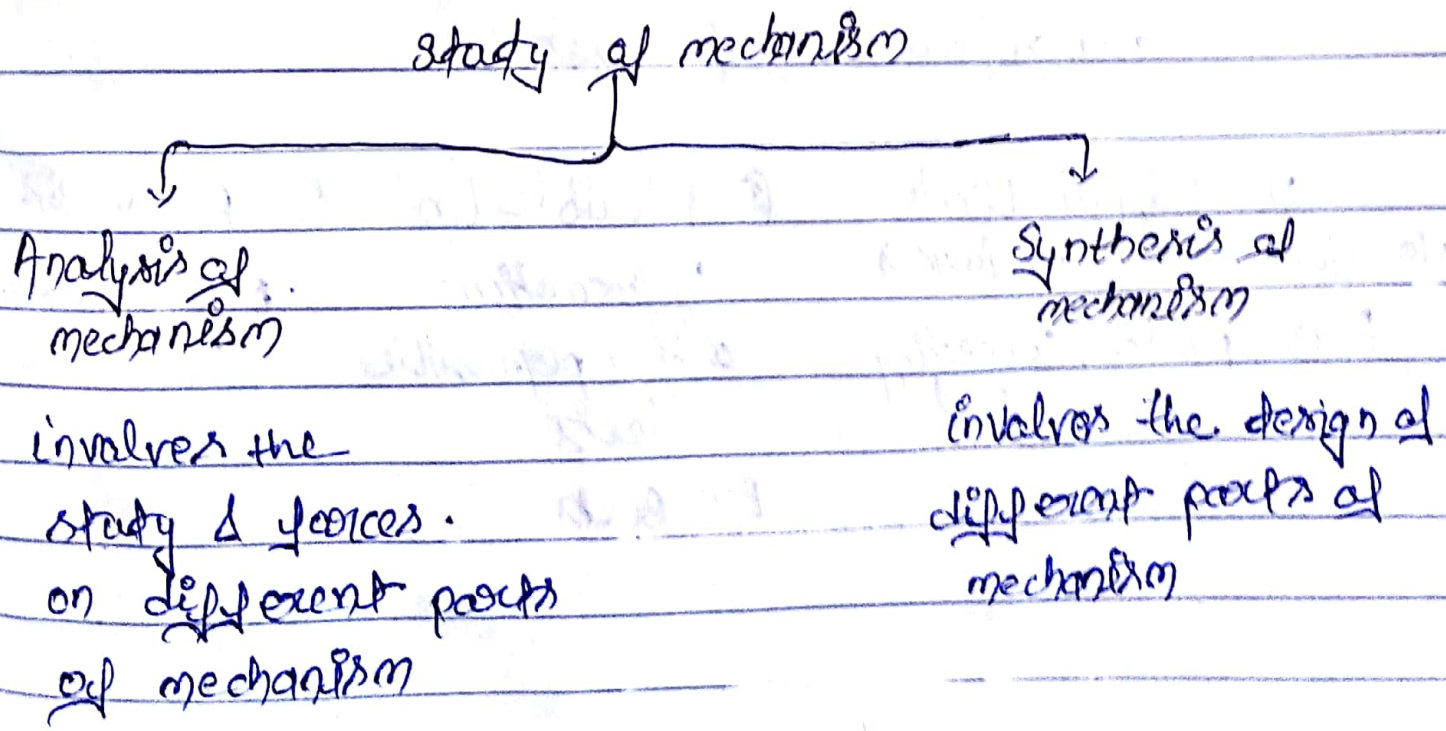


Mechanism:- No. of bodies assembled in such a way that the motion of one causes constrained and predictable motion to the other is known as a mechanism.



Machine:- combination of mechanisms which, apart from transmitting a definite motion, also transmit and modify available energy to work output.



"a resistant body with rigid connection preventing their movement is known as link"

* Link:- "a combination of members of mechanism, connecting other members and having motion relative to them."

→ Kinematic Link:- When a link has relative motion with any other link/element it is known as kinematic link/element.

Link

Binary

ternary

quaternary



these are the rigid link classification of link

① Rigid Link

deformation on links

Ex:- piston, connecting rod

② Flexible Link

deformation is within permissible limits

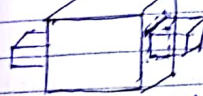
Ex:- Belts

③ Fluid Link

Ex:- Hydraulic Lift, Hydropneumatic

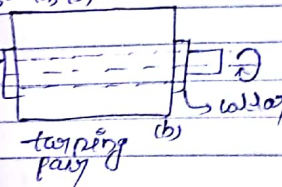
Types of constrained motion

1. Completely constrained motion: when the motion b/w two elements/link of a pair is in a definite direction



sliding pair (a)

Ex:- (a), (b)



turning pair (b)

{ motion may be linear or rotary }

2. Incompletely constrained motion: when the motion b/w two links/elements of a pair is possible in more than one direction.



Ex:-

3. Successfully constrained motion: when the motion b/w two links/elements of a pair is possible in more than one direction but is made to have motion only in one direction by some external means.



Ex:- Footstep bearing, valve in ic engine

Footstep bearing

{ Note:- Rigid body: a body is said to be rigid if ~~the~~ under the action of forces it does not suffer any distortion.

Resistant bodies: which are rigid for the purposes they have to serve. }

K. pair: kinematic pair or simply a pair is a joint of two links having relative motion to them.

Types of kinematic pairs

- * Nature of contact
- * Nature of mechanical constraint
- * Nature of relative motion

* * Nature of contact

(a) Lower pair: A pair of links having surface or area contact b/w the members is known as a lower pair. The contact surface of the two links are similar.

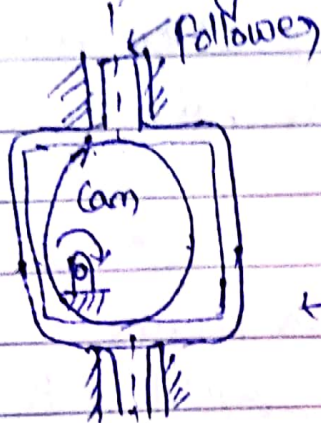
ex: Nut turning on a screw

(b) Higher pair: when a pair has a point or line contact b/w the links.

The contact surfaces of the two links are dissimilar.

ex: wheel rolling on a surface, cam and follower, tooth gear, ball and roller bearings etc.

* K. pair acc. to Nature of mechanical constraints
 (c) closed pair: when the elements of a pair are held together mechanically.

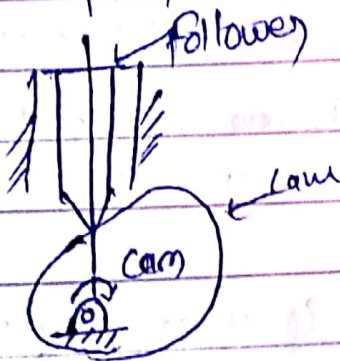


{ Note: All the lower pair and some of the higher pair are closed pairs. A cam and follower (higher pair)

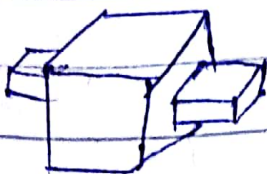
← Lower pair

(b)

Unclosed pair: when two links of a pair are in contact either due to force of gravity or some spring action, they constitute an unclosed pair.

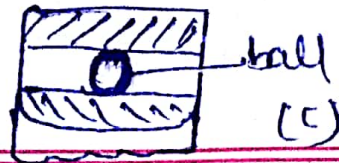


* K. pairs acc. to Nature of R. Motion
 (a) Sliding pair: of two links having a sliding motion to each other

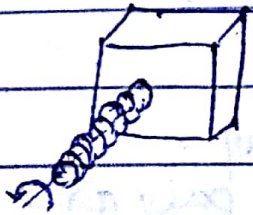


(b) Turning pair: when one link has a turning or revolving motion relative to other.





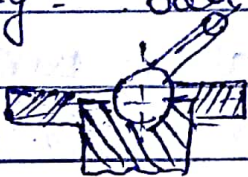
(c) Rolling Pair: when the links of a pair have a rolling motion relative to each other.



(d) Screw pair: If two mating links have a turning as well as sliding motion b/w them

(e) Spherical pair: when one link in the form of a sphere turns inside a forced link.

e.g.: ball and socket joint

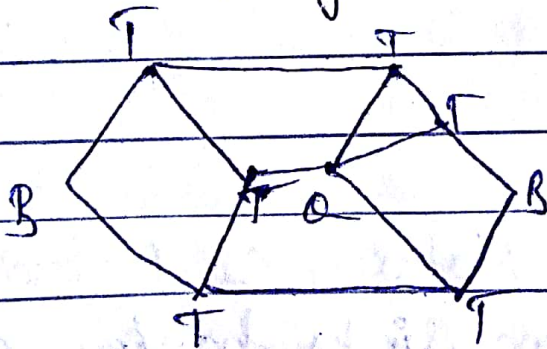


* Types of joints

* Binary joint: If two links are joined at the same corner

* Ternary " " " three links are joined at a corner

* Quaternary " " " four " " " " " "

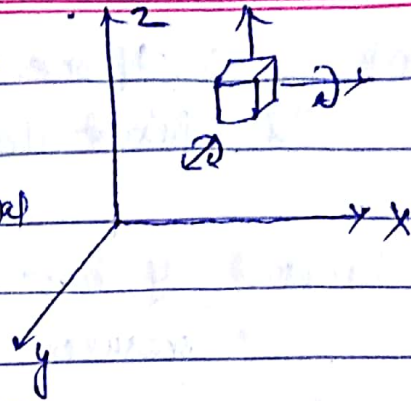


states if n no. of links are connected at a joint it is

equivalent to $(n-1)$ binary joints

"DOF"

DOF: DOF of a pair is defined as the number of independent relative motions, both translational and rotational

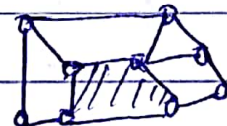
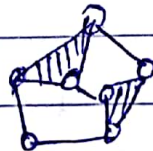
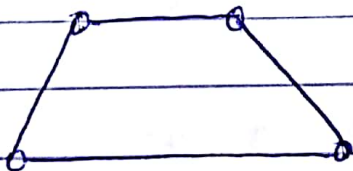


- * a rigid body possesses 6 DOF
- * The no. of constraints can never be zero or six

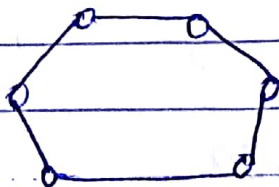
$$\text{DOF} = 6 - \text{Number of constraints}$$

* Kinematic chain *

A kinematic chain is an assembly of links in which the relative motion of the links is possible and the motion of each relative to the other is definite.



* Non kinematic chain: motion of a link results in indefinite motion of other link.



* A redundant chain does not allow any motion of a link relative to the other:



Note: * Dof of a locked system is zero
 * A structure with -ve dof is known as superstructure

* Linkage: If one of the links of a kinematic chain is fixed to the ground.

* Mechanism: If motion of any of the movable links results in definite motions of the other.

* Structure: If one of the links of a redundant chain is fixed, it is known as structure of a locked system.

* Gruebler's ~~equation~~ formula

$$F = 3(N-1) - 2P_1 - P_2$$

$$F = \text{Dof}$$

N = Total no of link

P_1 = No. of pairs having one dof

P_2 = " " " " two dof

($N-1$) one link is fixed.

* Kutzbach's criterion

$$F = 3(N-1) - 2P_1 - P_2$$

where $P_2 \geq 0$

When number of loops are known in a kinematic chain

$$F = N - (2L + 1)$$

$$P_1 = N + (L - 1)$$

Note: 1. If shortest link & longest link is
sum of shortest and longest link
is ~~greater~~ ^{less} than other two

then it is a mechanism of case-1

* If shortest link is fixed then it is
crank - crank or double crank or drag-crank
or rotatory-rotatory ~~other~~ converter.

2. If the link adjacent to the shortest link is
fixed then it is double rocker or double lever or
rocker-2 or oscillation-2 converter.

* If sum of shortest & longest link is greater
than other two then it is case-II four bar
linkage.

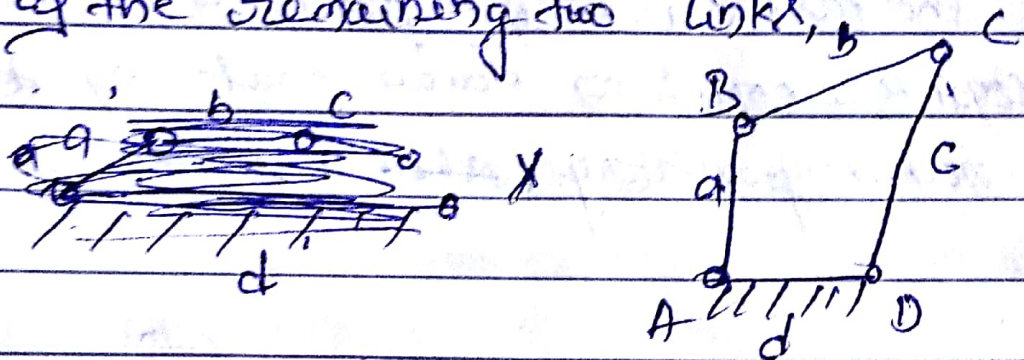
* If link app. to the shortest link is fixed = Double ~~crank~~ ^{lever}

* If the shortest link is fixed = Double crank

* If the link adjacent to the shortest
link is fixed = crank-lever

Graashof's criterion

Graashof's law:- it states that for a four-bar linkage system, the sum of the shortest and longest link of a planar quadrilateral linkage is less than or equal to the sum of the remaining two links,



$$a + d < b + c$$

* A link that makes complete revolution is called the crank.

* link opposite to the fixed link is called coupler

* and 4th link is called lever or rocker

if it oscillates then rocker or lever

if rotates then another crank.

* Mechanical Advantages *

power input = power output

$$MA = \frac{T_4}{T_2} = \frac{\omega_2}{\omega_4}$$

* Transmission Angle (μ) (angle b/w coupler and output link)

Applying cosine law to Δ 's ABD and BCD

~~$$a^2 + d^2 - 2ad \cos \mu = k^2$$~~

$$a^2 + d^2 - 2ad \cos \mu = k^2$$

$$b^2 + c^2 - 2bc \cos \mu = k^2$$

$$a^2 + d^2 - 2ad \cos \mu = b^2 + c^2 - 2bc \cos \mu$$

$$\frac{d\mu}{d\theta} = 0 \quad \text{for min \& max value}$$

$$\frac{d\mu}{d\theta} = \frac{ad \sin \mu}{bc \sin \mu}$$

$$\frac{d\mu}{d\theta} = 0$$

$$\therefore \theta = 0^\circ \text{ then } \mu_{\min}$$

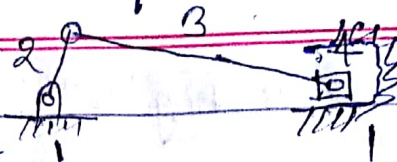
$$\theta = 180^\circ \text{ then } \mu_{\max}$$

The slider-crank chain / single slider crank mechanism

1st Inversion

Cylinder is fixed

- uses:
- 1. Reciprocating engine 4 (Piston) is driver
 - 2. Reciprocating compressor 2 (Crank) is driver

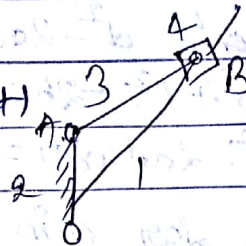


2nd Inversion

(Crank is fixed)

uses: 1. W. O. R. O. M

2. Rotating engine

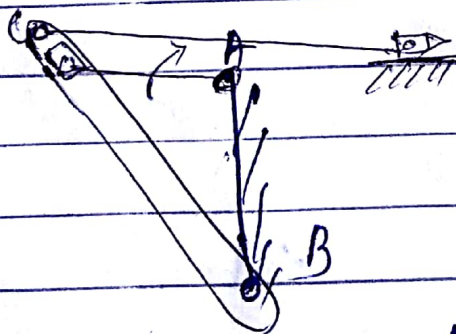


Note

{ W. O. R. O. M - It is a mechanism used in workshop to cut the metals. The forward stroke takes a little longer and cuts the metal whereas the return stroke is little and takes a shorter period } (Whitworth quick return mechanism)

3rd Inversion & Connecting rod is fixed

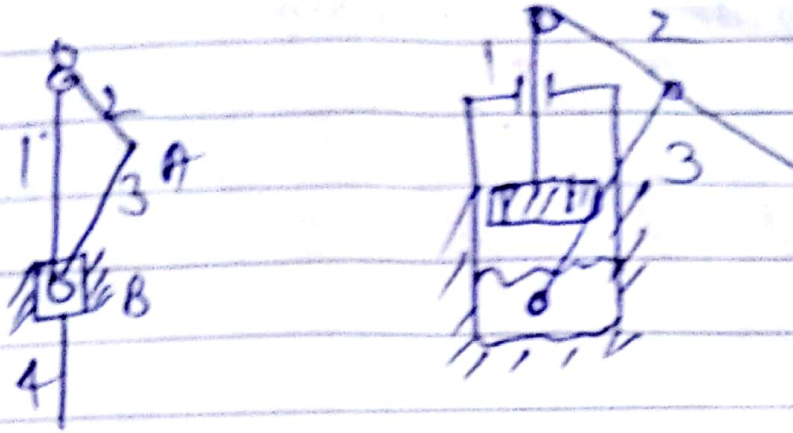
- uses:
- 1. Oscillating cylinder engine
 - 2. Crank and slotted-lever mechanism



$\frac{\theta}{\beta}$ = Quick Return Ratio

{ θ = cutting stroke angle
 β = Return stroke angle }

1st Inversion (Slider is fixed)



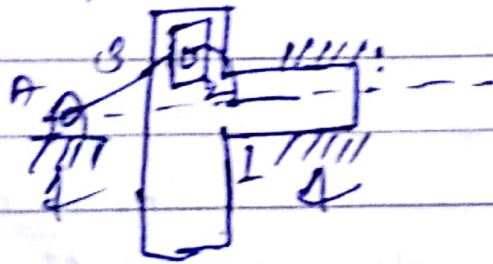
App: Hand pump.

Double slider - crank chain

1st Inversion

App: Elliptical Trammel

2nd Inversion



App: Scotch yoke

3rd Inversion

App: Oldham's coupling

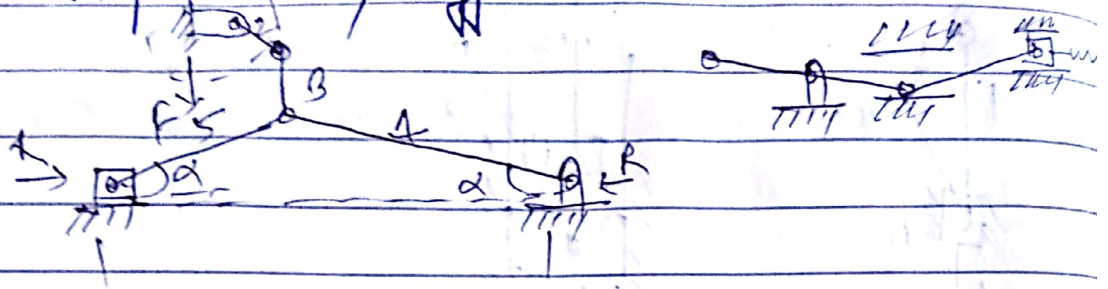
Note :

$$\omega = \frac{2\pi N}{60}$$

maximum sliding velocity = $\omega \times d$

* Snap-Action Mechanism

the mechanism used to overcome a large resistance of a member with a small driving force are known as snap action or toggle mechanism.



* Indexing Mechanism

Indexing mechanism serves the purpose of dividing the periphery of a circular piece into a number of equal parts.

* BR-2P, 2R-2P (Revalute-prismatic)

Inversion $\rightarrow$
 2R-2P $\rightarrow$ application $\rightarrow$ elliptical cammel (1st)
 (RRPP or RPRP) $\rightarrow$ 1) Scotch yoke (2nd)
 2) Oldham's coupling (3rd)

Inversion
 BR-P $\rightarrow$ Application $\rightarrow$ 1st
 (RRPR or RPRR) Reciprocating engine
 (RRRP) " Compressor

2nd inversion
 * QRM $\rightarrow$ shaper
 3rd inversion
 * oscillating
 * crank and
 (shaping)